



Vishaali B 251001304-2025-IT ▾

V2

Started on	Saturday, 15 August 2026, 10:59 PM
State	Finished
Completed on	Friday, 21 August 2026, 1:59 PM
Time taken	5 days 14 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int num;
6      int c=0;
7      c++;
8      scanf("%d", &num);
9
10     for (int i = 1; i <= num; i++)
11     {
12         c++;
13         c++;
14         if (num % i == 0)
15         {
16             c++;
17         }
18     }
19
20
21
22
23     printf("%d", c);
24
25
26 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)